Continuous-Time Reinforcement Learning:

A Literature Review

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1 Introduction

Reinforcement learning (RL) has traditionally been formulated in discrete time, where an agent observes a state, selects an action, and receives a reward at a sequence of distinct time steps. While this framework is natural for many settings such as turn-based games or episodic decision problems, many systems evolve continuously in time. Robotic locomotion, financial portfolio management, physiological regulation, and autonomous vehicle navigation are often better described by differential equations, i.e., a continuous framework, than by discrete-time recursions. These considerations have motivated the development of continuous-time reinforcement learning, which seeks to frame sequential decision-making problems using tools from stochastic processes, differential equations, and control theory.

Moving from discrete to continuous time introduces complexity, though not without opportunities. On the modeling side, the Hamilton–Jacobi–Bellman (HJB) equation replaces the Bellman recursion, stochastic differential equations (SDEs) take the place of Markov chains, and pathwise calculus provides a natural framework for analyzing trajectories. At the same time, several challenges arise. Function approximation becomes more delicate, gradient estimators that behave well in discrete time can exhibit large variance in the continuous limit, and the optimization procedures used to train value functions and policies are themselves

stochastic dynamical systems whose behavior shapes what is learned. The literature reviewed here addresses these issues from several, ideally complementary, angles.

A useful lens for interpreting the literature surveyed below is the systematic replacement of discrete-time constructs with objects native to continuous-time dynamics. In each major thread, a quantity inherited from the discrete-time formulation—such as a TD error, a likelihood-ratio gradient, or an L^2-projected Bellman operator—is shown to exhibit pathological behavior as the discretization step shrinks. The resulting continuous-time methods reformulate these quantities in terms of objects compatible with diffusion dynamics, including HJB residuals, pathwise derivatives, and Sobolev-proximal operators. The progression from Doya [2000] through Munos [2006] to Mou [2026] can be read as successive instantiations of this substitution.

The goal of this review is to provide a structural understanding of how continuous-time dynamics reshape the reinforcement learning problem, with an example utilizing those dynamics to show connections between concepts. That example, specifically in a financial context, in Section 6 is treated in greater depth than the preceding sections, serving to connect the theoretical machinery developed earlier to a concrete problem domain where those ideas find natural expression. This document is organized into five sections: (1) foundational formulations and actor–critic methods in continuous time; (2) policy gradient estimation in the continuous-time limit; (3) theoretical guarantees for RL with function approximation; (4) enabling tools, including continuous-time sequence models; and (5) an application to portfolio optimization under partial information.

2 A Quick Primer: Core Concepts

Concepts: Discrete- and Continuous-Time Foundations

Discrete-Time Foundations

State Space, Actions, and the Markov Decision Process. A discrete-time **Markov Decision Process** (MDP) is a tuple $(\mathcal{S}, \mathcal{A}, P, r, \gamma)$. The **state space** $\mathcal{S}$ collects every configuration the environment can occupy; at each time step t the agent observes $s_t \in \mathcal{S}$ and selects an action $a_t \in \mathcal{A}$. The environment then transitions stochastically according to the **transition kernel**

$$P(s_{t+1} \mid s_t, a_t),$$

which satisfies the Markov property: The next state depends only on the current state–action pair, not on prior history.

Reward and Cost Functions. Reinforcement learning problems can be framed equivalently in terms of *rewards* (to be maximised) or *costs* (to be minimised); the two are related by a sign flip and both appear in the literature. The **reward function** $r : \mathcal{S} \times \mathcal{A} \to R$ assigns a scalar signal received immediately after taking action a_t in state s_t, and the agent seeks a policy $\pi : \mathcal{S} \to \mathcal{A}$ maximising the *expected discounted cumulative reward*

$$J(\pi) = E_\pi \left[\sum_{t=0}^{\infty} \gamma^t \, r(s_t, a_t) \right], \quad \gamma \in [0, 1).$$

In control-oriented treatments a non-negative **cost function** $c : \mathcal{S} \times \mathcal{A} \to R_{\geq 0}$ replaces the reward and the objective becomes minimisation; this framing is especially prevalent in continuous-time and optimal control settings, where c often takes the quadratic form $c(s, a) = s^\top Q s + a^\top R a$. The discount factor γ serves a dual role, as it ensures the infinite sum converges and encodes a preference (characterized as a greedy policy) for near-term outcomes over distant ones.

Value Functions. Given a policy π, the **state-value function** $V^\pi : \mathcal{S} \to R$ measures

expected return from each state,

$$V^\pi(s) = E_\pi\left[\sum_{k=0}^{\infty} \gamma^k\, r(s_{t+k}, a_{t+k}) \;\middle|\; s_t = s\right],$$

while the **action-value function** (Q-function) $Q^\pi : \mathcal{S} \times \mathcal{A} \to R$ additionally conditions on the initial action:

$$Q^\pi(s,a) = E_\pi\left[\sum_{k=0}^{\infty} \gamma^k\, r(s_{t+k}, a_{t+k}) \;\middle|\; s_t = s,\; a_t = a\right].$$

These are commonly known as the Bellman Equations. Combining these two equations, the **advantage function** $A^\pi(s,a) = Q^\pi(s,a) - V^\pi(s)$ isolates the relative benefit of action a over the average behaviour of π at s, and plays a central role in the actor–critic methods developed in the body text of this section.

Backward Euler Approximation, Eligibility Traces, and TD Methods. A natural numerical handle on the value function is the **backward Euler** finite difference $\dot{V}(s_t) \approx \big(V(s_t) - V(s_{t-1})\big)/\Delta t$, which trades the exact continuous-time update for a stable implicit recursion and, in RL, yields the familiar **TD(0)** (temporal difference) update toward the one-step Bellman target $r_t + \gamma V(s_{t+1})$. **Eligibility traces** extend this by assigning a decaying credit $e_t(s) = \gamma\lambda\, e_{t-1}(s) + \mathbf{1}[s_t = s]$ to each state, reflecting how *recently* and *frequently* it was visited. The TD error $\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t)$ is then broadcast to all previously visited states weighted by their traces,

$$V(s) \;\leftarrow\; V(s) + \alpha\, \delta_t\, e_t(s),$$

interpolating between pure one-step TD ($\lambda = 0$) and full Monte Carlo returns ($\lambda = 1$); this family is collectively denoted **TD(λ)**.

Continuous-Time Extensions

Hamilton–Jacobi–Bellman (HJB) Equation. For a controlled diffusion $dX_t = f(X_t, a_t)\,dt + \sigma(X_t)\,dW_t$ with running reward $r(x, a)$, the HJB equation characterizes the optimal value function $V^*(t, x)$ as the solution to the PDE

$$-\partial_t V(t,x) \;=\; \max_a \Big[r(x,a) + \nabla_x V(t,x)^\top f(x,a) + \tfrac{1}{2}\,\mathrm{tr}\big(\sigma\sigma^\top \nabla^2_{xx} V(t,x)\big)\Big].$$

The HJB Equation replaces the discrete-time Bellman recursion: the maximisation over actions and the diffusion term $\frac{1}{2}\,\mathrm{tr}(\sigma\sigma^\top \nabla^2_{xx} V)$ together encode both the optimality condition and the effect of stochastic noise on value. The continuous-time TD error (discussed later in Section 2) corresponds to evaluating the residual of this equation along observed trajectories, replacing the unknown value function derivatives with estimates obtained from sampled paths.

A remark on sample paths is worth making before turning to the algorithms. The HJB equation is fundamentally a *distributional* result: It characterizes V^* through expectations over the process, and a learning agent never observes a distribution—it observes a single sample path $\omega \mapsto X_t(\omega)$. Evaluating the HJB residual therefore means substituting $\nabla_x V$ and $\nabla^2_{xx} V$ with estimates computed along that specific trajectory, not in expectation. The pathwise residual is an unbiased estimator of the true HJB error by the law of total expectation, so distributional correctness is preserved on average; but the individual path also carries trajectory-level variability that aggregate summaries suppress. This is precisely what eligibility traces and martingale-based policy evaluation exploit, as both assign credit by tracking accumulated discrepancy along one observed path, relying on the fact that under the optimal policy the process $V(t, X_t) + \int_0^t r(s, X_s)\,ds$ is a martingale *pathwise*, not merely mean-zero in distribution.

One of the earliest systematic treatments of continuous-time reinforcement learning is Doya [2000], who formulates the learning problem through the HJB equation for infinite-

horizon discounted reward. Rather than deriving the continuous setting as a limiting case of discrete-time RL, Doya develops learning updates directly from a continuous-time temporal difference (TD) error defined as the instantaneous discrepancy between the rate of change of the value function and the observed reward rate. From this formulation, both backward Euler approximations and eligibility-trace variants are derived, which correspond closely to the TD(0) and TD(λ) algorithms described at the beginning of this section.

A key contribution of Doya [2000] is the presentation of two related policy improvement strategies. The first is a continuous-time **actor–critic** method, whose two components play distinct roles: the **critic** maintains and updates the value function $V(x)$, tracking the HJB residual along observed trajectories to produce a running estimate of how good the current state is; the **actor** then uses the TD error $\delta(t)$ as a reinforcement signal to update the policy parameters, stepping in directions the critic signals as beneficial. The actor does not act on raw rewards directly (because it receives only the TD error), which folds the reward into the critic's running value estimate and thereby summarizes whether the current state trajectory is better or worse than expected. The second approach derives a value-gradient-based greedy policy that admits a closed-form feedback control law when the system dynamics are linear in the control input, replacing the actor network with an analytic solution that reads off the optimal action directly from the gradient $\nabla_x V$. Experiments on pendulum and cart–pole swing-up tasks indicate that the continuous actor–critic method can require fewer trials than comparable discrete-time approaches, while value-gradient policies based on learned dynamics can be substantially more sample efficient. These results established a basic framework that has influenced much of the subsequent work on continuous-time RL, and the HJB residual as a continuous-time TD error remains a central computational object in the literature that follows.

3 Policy Gradient Estimation in Continuous Time

Once value-based methods are available, a natural question arises concerning the construction of reliable policy gradient estimators. In discrete time, the likelihood ratio (or score function) estimator provides an unbiased estimate of the gradient of expected return with respect to policy parameters. For a parameterized policy π_θ, this estimator takes the form $\nabla_\theta \log \pi_\theta(a \mid s)$, the gradient of the log-likelihood of the chosen action. However, Munos [2006] shows that this approach behaves poorly in the continuous-time setting. As the discretization step $\Delta \to 0$, the number of decision points over a fixed horizon grows as $N = T/\Delta$. The likelihood-ratio estimator accumulates a sum of N stochastic score terms, and its variance therefore scales linearly with the number of decisions, diverging as the time grid is refined. Consequently, naive score-function methods become impractical for finely discretized control problems. This is a second instance of the theme identified in the introduction: a discrete-time estimator that is well-behaved at fixed Δ breaks down in the limit.

To address this issue, Munos proposes a pathwise derivative estimator. The key observation is that the policy gradient can be expressed in terms of a state-gradient process that evolves alongside the system trajectory. If the system dynamics were known, this Jacobian process would allow direct computation of the gradient without the variance explosion associated with likelihood ratios. In practice, the method introduces a stochastic policy so that the system can be treated as a stochastic process, allowing unknown derivatives of the dynamics to be replaced by likelihood ratio terms computed from the policy. The remaining Jacobian terms are then estimated using local least-squares regression on observed state transitions.

Under suitable conditions, i.e., regularity conditions for this setting, the resulting estimator converges almost surely to the true policy gradient as $\Delta \to 0$, with variance that *decreases* rather than increases in the limit. Munos provides empirical validation on a six-dimensional spring–mass system and a double inverted pendulum, demonstrating convergence to near-optimal controllers in both cases. Together, the works of Doya [2000] and Munos [2006]

highlight a key structural distinction in continuous-time RL: value-based methods naturally exploit the geometry of the HJB equation, while policy-gradient methods require gradient estimation strategies that are explicitly designed for the continuous limit rather than inherited from discrete time. The takeaway is that the continuous-time setting does not merely complicate existing estimators, it calls for new estimators using objects and concepts native to diffusion dynamics.

4 Theoretical Guarantees: Function Approximation and Statistical Learning

Concepts: Norms, Stability, and the Deadly Triad

NORMS AND STABILITY

Supremum Norm ($\|\cdot\|_\infty$). For a bounded function $f : \mathcal{S} \to R$, the supremum norm is $\|f\|_\infty = \sup_{s\in\mathcal{S}} |f(s)|$. It measures the worst-case deviation across all states uniformly. The Bellman operator $\mathcal{T}$ is a γ-contraction under this norm, $\|\mathcal{T}V - \mathcal{T}U\|_\infty \leq \gamma \|V - U\|_\infty$, guaranteeing convergence of exact (tabular) value iteration.

L^2 Norm ($\|\cdot\|_\mu$). For function approximation one typically minimizes a weighted squared error, $\|f\|_\mu^2 = \int f(s)^2 \, d\mu(s)$, where μ is a state-visitation distribution. Because the Bellman operator is not generally a contraction in this norm, projecting onto a function-approximation class can expand rather than shrink errors.

The Deadly Triad. Sutton and Barto [2018] identify three individually benign ingredients that together cause divergence in value-function methods:

1. **Function approximation**: representing V or Q with a parameterized class (e.g. linear features, neural networks) rather than a lookup table.

2. **Bootstrapping**: updating estimates using other estimates as targets (as in TD and DP methods), rather than waiting for full Monte Carlo returns.

3. **Off-policy learning**: training on a data distribution μ that differs from the on-policy distribution d^π, breaking the contraction properties that tabular TD enjoys.

Any two of the three can coexist stably, but all three simultaneously can produce unbounded value estimates even for simple linear approximators. The core tension is that the L^2 projection and the Bellman operator are each non-expansive individually, but their composition can have spectral radius greater than one under an off-policy μ.

Ellipticity and the Sobolev Norm

Uniform Ellipticity. A controlled diffusion with diffusion coefficient $\sigma(x)$ is said to satisfy a uniform ellipticity condition if there exists $\lambda > 0$ such that

$$\xi^\top \sigma(x)\sigma(x)^\top \xi \;\geq\; \lambda \|\xi\|^2 \quad \text{for all } x \in \mathcal{S},\ \xi \in R^d.$$

Intuitively, this requires the noise to inject perturbations in every direction uniformly, preventing the process from becoming deterministic along any coordinate. It is the stochastic analogue of strict positivity of the diffusion matrix.

Sobolev Norm ($\|\cdot\|_{H^1}$). Where the L^2 norm $\|\cdot\|_\mu$ measures the size of a function, the Sobolev norm additionally penalizes the size of its gradient:

$$\|f\|^2_{H^1(\mu)} \;=\; \int \left[f(s)^2 + \|\nabla f(s)\|^2\right] d\mu(s).$$

Performing the Bellman projection under this norm rather than the plain L^2 norm incorporates gradient information and, under uniform ellipticity, yields a positive-definite operator. This breaks the instability issue at the heart of the deadly triad.

A longstanding challenge in reinforcement learning concerns the behavior of value function approximation. In discrete-time RL, the Bellman operator is a contraction in the supre-

mum norm but not in the L^2 norm typically used for function approximation, contributing to the deadly triad of instability described at the start of this section. Mou [2026] examines this issue in the context of continuous-time diffusion processes and argues that the ellipticity intrinsic to Markov diffusions induces positive-definite structure in Sobolev spaces that resolves the instability—making value function learning in this setting more tractable than the general discrete-time case.

The central observation of Mou [2026] is that controlled Markov diffusions satisfying the uniform ellipticity condition possess additional regularity properties that induce a positive-definite structure in Sobolev spaces. This is the continuous-time analogue of the theme running through the earlier sections: Ellipticity is a structural feature native to diffusion processes that has no counterpart in discrete Markov chains, and exploiting it changes the character of the learning problem from potentially unstable to well-posed. The paper proposes a *Sobolev-prox fitted q-learning* algorithm for the off-policy setting, in which offline data collected under a fixed behavior policy is used to learn the optimal value and advantage functions without knowledge of the system dynamics. The value function and advantage function are estimated separately, with the value update performed through a proximal step under a Sobolev norm rather than an L^2 projection.

The main theoretical results provide oracle inequalities decomposing the estimation error into four components: (1) approximation error within the chosen function classes, (2) statistical complexity terms characterized by critical radii, (3) optimization error that decays geometrically with iterations, and (4) numerical discretization error arising from the continuous-to-discrete approximation. Importantly, these guarantees rely primarily on ellipticity and mild smoothness conditions on the drift and diffusion coefficients, rather than strong assumptions about data coverage or Bellman completeness. One implication is that, in this diffusion setting, value-function estimation resembles a well-posed statistical learning problem rather than an unstable fixed-point iteration, with error bounds governed by approximation, estimation, and optimization terms in a manner analogous to nonparametric

regression.

Before proceeding, it is worth acknowledging two structural limitations of the continuous-time framework as developed above. First, the theoretical guarantees—particularly those of Mou [2026]—rely on assumptions that may not hold in practice. The uniform ellipticity assumption requires noise in every coordinate direction, and the smoothness conditions on drift and diffusion coefficients exclude many models used in certain applications. Second, there is an inherent tension between continuous-time theory and discretely observed data: Real systems are sampled at finite intervals, and the continuous limit is always an approximation whose quality depends on the sampling rate and the smoothness of the underlying process. However, these caveats do not undermine the value of the framework, but they do mean that the methods discussed here rely on different and not always overlapping assumptions. That is to say: Pathwise gradient estimators and Sobolev-proximal updates address different failure modes and are not interchangeable.

Because value functions in continuous-time RL are typically estimated via stochastic gradient methods, the optimization procedure itself induces a stochastic dynamical system whose behavior influences the learning process. Hu et al. [2018] and Li et al. [2017] show rigorously that SGD on nonconvex objectives can be approximated by a stochastic differential equation. Hu et al. [2018] further applies large deviation theory and exit-time analysis to the approximating diffusion, suggesting that escape from a local minimizer requires exponentially many iterations in η^{-1}, while escape from a saddle point requires only $O(\eta^{-1} \log \eta^{-1})$ iterations (though the authors note these rates are properties of the continuous diffusion approximation rather than direct proofs for the discrete SGD).

The SGD-as-SDE perspective closes out the theoretical core of this review. A separate but equally practical challenge arises from the data side: continuous-time RL applications often involve observations that arrive asynchronously or with gaps, and standard sequence models are poorly suited to this setting. The following section describes a class of models designed specifically for irregularly sampled time series, which will reappear as a natural tool

in the finance application in Section 6.

5 Enabling Infrastructure: Neural Controlled Differential Equations

Continuous-time RL requires not only sound algorithms and theoretical guarantees, but also state representations that can be constructed from data arriving asynchronously or with missing observations—a challenge that becomes acute in partially observed environments such as financial applications. Neural Controlled Differential Equations (CDEs) function as state representation tools in the partially observed setting: They process irregularly sampled observation paths to produce hidden-state sequences suitable for downstream policy and value learning, but do not themselves solve the control problem.

Many continuous-time RL applications therefore require models capable of representing states derived from irregularly sampled time series. Kidger et al. [2020] introduce Neural CDEs, a class of continuous-time sequence models designed to address this problem. Unlike Neural ODEs, whose trajectories are fully determined once the initial condition is fixed, Neural CDEs allow the hidden state to evolve in response to a continuous interpolation of the observed data. Given a spline interpolation $X : [t_0, t_n] \to R^{v+1}$ of the observations, the hidden state evolves according to

$$z_t = z_{t_0} + \int_{t_0}^{t} f_\theta(z_s)\, dX_s, \tag{1}$$

making the hidden dynamics responsive to the incoming observation path rather than evolving autonomously from an initial condition alone.

This formulation can be viewed as the continuous-time analogue of a recurrent neural network, in the same sense that Neural ODEs correspond to continuous versions of residual networks. Several practical advantages follow. From such a formulation, Neural CDEs

handle irregular sampling and missing observations naturally, avoid repeated solver interruptions used in earlier ODE–RNN hybrids, and allow efficient adjoint-based backpropagation across the full trajectory. Empirical results show competitive performance on tasks including CharacterTrajectories, PhysioNet sepsis prediction, and Speech Commands classification [Kidger et al., 2020]. For continuous-time RL, such models provide a flexible mechanism for constructing state representations when observations arrive asynchronously—a situation that arises naturally in partially observed environments and in financial applications where market data arrives at irregular intervals. Importantly, Neural CDEs address the state representation problem upstream of the control problem; integrating them with HJB-based value learning under partial observability remains an open challenge, as discussed in Section 6.

6 Application to Finance: Portfolio Selection under Partial Information

The works surveyed above share a common structural logic: each begins with a construct that is natural in discrete time and examines its behavior as the discretization step Δ tends to zero. When the limiting behavior becomes pathological, the method is reformulated using objects that are intrinsic to continuous-time dynamics. Doya's continuous-time TD error is the HJB residual, not a finite-difference approximation to it. Munos's pathwise derivative estimator replaces a score-function gradient whose variance diverges in the limit. Mou's Sobolev-proximal update replaces an L^2 projection whose composition with the Bellman operator can be explosive. In each case, the key insight is that continuous time is not a limit to be approached from discrete time, but a setting with its own geometry—the geometry of diffusion operators, Sobolev spaces, and the HJB PDE—that can and should be exploited directly.

Continuous-time portfolio optimization provides one of the most demanding testbeds for the theoretical machinery surveyed above, and the range of RL approaches now applied

to it is broad [Hambly et al., 2023]. The discussion here is intended to be illustrative rather than exhaustive; the treatment focuses on the structural features of the algorithm most directly connected to the theoretical machinery developed, as noted in this review. Among recent contributions, Wu and Li [2024] develop an actor–critic algorithm for mean-variance (MV) portfolio selection in a regime-switching market with an unobservable regime. The setting directly instantiates the HJB-centered framework of Doya while also confronting two challenges that were identified as open frontiers in the preceding sections, i.e., the issues of partial observability and score-function instability in the continuous-time limit.

To fix the portfolio objective precisely before describing the algorithm, recall the Mean Variance Portfolio Theory of Markowitz [1952]. Consider the return on an investable asset as $r = \frac{x_t - x_0}{x_0}$, where x_t is the sale price after holding for time t and x_0 is the purchase price. Let r_i be the random variable associated with the rate of return of asset i, for $i = 1, 2, \ldots, n$, and let $z = (r_1, r_2, \ldots, r_n)^\top$ be the random vector of returns.

Set $\mu_i = E(r_i)$, $m = (\mu_1, \mu_2, \ldots, \mu_n)^\top$, and $\mathrm{cov}(z) = \Sigma$. Let $w = (w_1, w_2, \ldots, w_n)^\top$ be a vector of portfolio weights. The portfolio return $r_p = w^\top z$ is a random variable with mean $m^\top w$ and variance $w^\top \Sigma w$. Let μ_b be the acceptable baseline expected rate of return. The central optimization problem of Mean-Variance Portfolio Theory is then

$$\begin{aligned} \min_{w} \quad & \frac{1}{2} w^\top \Sigma w \\ \text{subject to} \quad & m^\top w \geq \mu_b, \\ & \mathbf{1}^\top w = 1 \end{aligned}$$

Higher expected returns can only be achieved by accepting greater variance, and the structure of the covariance matrix Σ determines the extent to which diversification can mitigate this risk.

A key assumption underlying this formulation is covariance stationarity: the mean

vector m and covariance matrix Σ are treated as fixed over time. In practice, however, financial markets exhibit pronounced non-stationarity, with periods of differing volatility, correlation structure, and average returns.

A common modeling approach is to assume that returns evolve according to a *regime-switching* process, where the market alternates between a finite set of latent states (e.g., bull and bear markets). In each regime k, the return distribution is characterized by its own parameters $(m^{(k)}, \Sigma^{(k)})$. For example, a bull market is typically associated with higher expected returns and lower volatility, while a bear market exhibits lower or negative expected returns and increased variance, often accompanied by stronger correlations across assets.

Consequently, the mean–variance optimal portfolio is no longer static, but depends on the current market regime. Because the regime is unobservable, the investor cannot condition her policy on it directly. Wu and Li address this through the separation principle of Xiong and Zhou [2007]: The problem decomposes into two steps: a Wonham filtering step, which processes the observable price path to produce a posterior probability $p_t = \Pr(\text{regime} = \text{bull} \mid \mathcal{F}_t)$ evolving as an Itô diffusion, followed by a stochastic control problem in the augmented observable state (X_t, p_t). The Wonham filter is the classical optimal Bayesian filter for a continuous-time hidden Markov chain. The filtering step is itself a continuous-time operation, and consistent with the broader theme of this review, that such a continuous-time structure can and should be exploited rather than discretized away.

To encourage exploration, the authors adopt the exploratory stochastic control framework of Wang et al. [2020], augmenting the MV objective with an entropy regularization term weighted by a temperature parameter Λ. The resulting regime-switching exploratory MV (RSEMV) problem admits a semi-analytical solution: The optimal value function takes the form

$$V^*(t, x, p) \;=\; (x - \omega)^2 \frac{1}{f(t, p)} + g(t, p) - (\omega - z)^2, \tag{2}$$

where f and g satisfy a pair of linear parabolic PDEs in (t, p), and the optimal explo-

ration policy is Gaussian with mean depending on both the current wealth deviation $(x-\omega)$ and the belief state p_t. This belief-state dependence is the key structural departure from the geometric Brownian motion case: regime uncertainty introduces a hedging demand term proportional to $\partial_p \ln f(t,p)$ that modulates risky asset exposure according to the investor's current regime estimate. When p_t is close to one-half (indicating maximum regime uncertainty) this hedging demand is largest, and the algorithm naturally reduces exposure to the risky asset.

Because environment parameters are unknown, the PDEs for f and g cannot be solved directly. Instead, Wu and Li approximate $\ln f(t,p)$ and $g(t,p)$ by low-order polynomials in p and $(T-t)$, learning the coefficients through actor–critic updates. The policy evaluation step exploits the martingale characterization of Jia and Zhou [2022], applied to the MV setting by Wang and Zhou [2020]: under any fixed policy, the process $V(t,X_t)+\int_0^t r(s,X_s)\,ds$ is a martingale, so residual drift along observed trajectories signals a mis-specified value function. The resulting algorithm, POEMV, therefore alternates between minimizing this martingale loss to evaluate the current policy and a policy gradient step derived from the score function of the Gaussian policy.

By anchoring parametrization to the semi-analytical structure rather than a general neural network, the algorithm keeps the learned parameter count small and gradient estimates stable. This addresses the instability concern raised by Munos [2006] through a different mechanism than pathwise differentiation. Rather than redesigning the gradient estimator, the authors reduce the dimension of the estimation problem until score-function methods are again reliable. However, this approach carries its own structural cost. The polynomial parametrization works precisely because the value function's analytic form is known in advance; it is not a general-purpose algorithm but a structured one, and its performance relies on the correctness of that structural prior. Additionally, the separation of filtering and control into sequential rather than joint steps—estimating environment parameters first, then learning the policy—introduces a source of bias that is not addressed within the algorithm

itself. The result is a method that is model-aided rather than fully model-free, a distinction that matters when assumptions are violated or the pre-estimation step is inaccurate.

Empirically, Wu and Li evaluate POEMV on 30 years of daily S&P 500 data, training on the first 20 years and testing on the final 10. Against the EMV algorithm of Wang and Zhou [2020], which ignores regime switching entirely, and a classical plug-in control approach, POEMV achieves substantially lower out-of-sample variance of terminal wealth while meeting the target mean return. Under daily rebalancing, POEMV variance is roughly half that of EMV and several times smaller than the classical approach; under monthly rebalancing this gap exceeds an order of magnitude. Both comparisons show greater POEMV resilience around the March 2020 market dislocation.

This financial application of continuous-time Reinforcement Learning connects to several open questions, both broad and specific to its application. First, the Wonham filter introduces additional diffusivity in the belief state that regularizes the augmented control problem, suggesting a natural counterpart to the ellipticity-based regularity exploited by Mou [2026]; whether Sobolev-prox guarantees extend to this partially observed setting, where the augmented state space includes a diffusively evolving belief coordinate, is a technically non-trivial question and remains open. Second, the polynomial parametrization sidesteps the deadly triad but relies on structural knowledge of the value function; extending to richer model classes, including the Neural CDE representations of Kidger et al. [2020] for irregularly sampled price data, is a natural direction that would require confronting the partial observability–HJB interaction identified in Section 5. Finally—and perhaps most fundamentally—the filtering step requires pre-estimating environment parameters from historical data before learning begins, introducing a source of bias that the authors acknowledge. A fully model-free treatment that integrates filtering and learning jointly, rather than sequencing them, would require solving a coupled estimation problem in which the filter's posterior and the value function's parameters evolve together. This is technically challenging because the filter depends on transition probabilities that the value function also uses,

so joint optimization does not decompose cleanly.

Resolving this coupling remains a central open problem for continuous-time reinforcement learning in partially observed financial settings. A further set of questions concerns the domain assumptions embedded in the model itself. The two-regime structure fixes both the number of market states and their interpretation, whereas in practice the boundary between bull and bear conditions is neither sharp nor stable over time, and real markets may exhibit richer latent structure. Similarly, the framework treats portfolio management as allocation across a single risky asset, abstracting away the within-class granularity—across equity sectors, bond types such as municipal bonds or treasury bills, or alternative instruments—that practitioners must navigate. Incorporating this structure would require either higher-dimensional state and action spaces or hierarchical formulations, both of which interact non-trivially with the filtering and control problems already identified above.

Taken together, these threads leave three concrete open challenges for continuous-time reinforcement learning. The first is the integration of filtering and control. Partially observed settings, of which the regime-switching portfolio problem is one instance, require that posterior estimation and value learning proceed jointly rather than sequentially, a coupling that resists clean decomposition and for which no fully model-free continuous-time solution currently exists. The second is the extension of the ellipticity-based regularity theory of Mou [2026] to augmented state spaces that include belief coordinates or other diffusively evolving latent quantities, where the structural conditions supporting Sobolev-proximal guarantees must be re-examined. The third is scaling. Martingale-loss training and structured parametrizations provide stability precisely by constraining the function class, and combining that stability with richer representations—such as the Neural CDE architectures of Kidger et al. [2020]—without reintroducing the score-function variance identified by Munos [2006] remains an open algorithmic problem. Progress on these fronts would bring continuous-time reinforcement learning closer to a unified framework applicable across the scientific and engineering domains where continuous dynamics are the natural starting point.

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